



Dictionary Learning of Dynamical Systems with Optimal Transport

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New Trend in Statistical Learning

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Dynamical systems

Dynamical systems are backbone models of temporally evolving phenomena:

Modeling:

$$x_{t+1} = g(x_t)$$

discrete

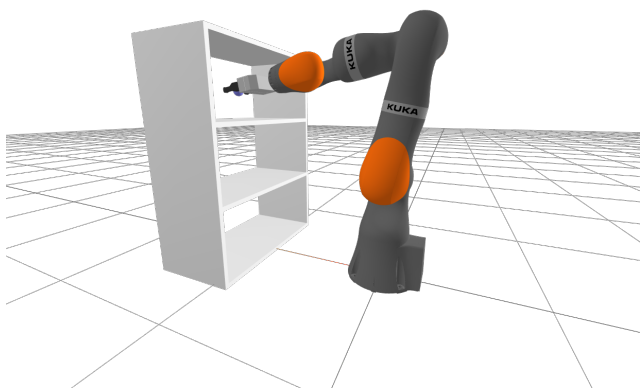
$$dx_t = g(x_t)$$

continuous

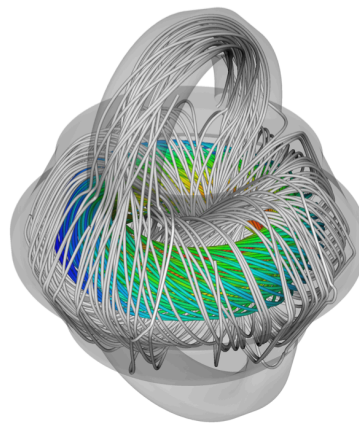
$$dX_t = g(X_t) + \sqrt{2\sigma}dB_t$$

stochastic

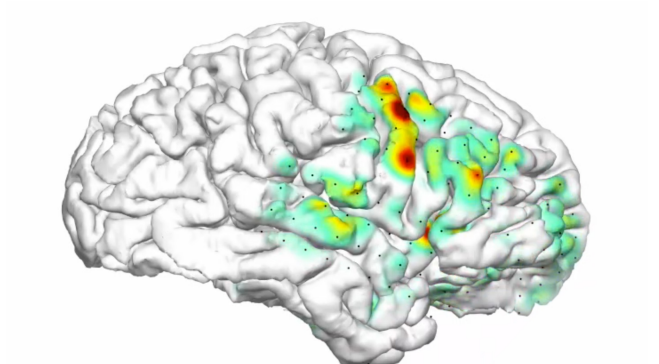
Application:



Robotic



Fluid dynamic



Brain dynamic

Machine learning for dynamical systems

Surrogate model & forecasting:

- ➡ Learning efficient models to predict future states (Neural operators).

Control & decision making:

- ➡ Understanding how interventions on state variables propagate through the system (causal inference).
- ➡ Learning control policies to drive system towards an objective (model-based RL)

System identification & model discovery:

- ➡ Learning the underlying transition equations (neuralODE)
- ➡ Learning coordinate systems in which dynamics become simple (Koopman theory)



Current methods typically consider a single dynamic.

- ➡ Necessity of long trajectory (large dataset)

Can we leverage knowledge about « similar » dynamics to perform system identification ?

- 1. Operator representations of dynamical systems**
- 2. Optimal Transport metric between operator representations**
- 3. Dictionary Learning for Dynamical Systems**

Operator representations of dynamical systems

Linear autonomous dynamical systems

Modeling:

$$\dot{x}(t) = Ax(t) \rightarrow x(t) = S_t(x_0) = \exp(tA)x_0$$

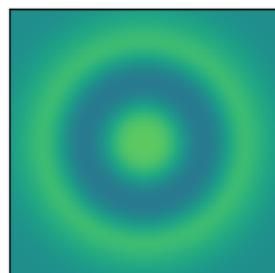
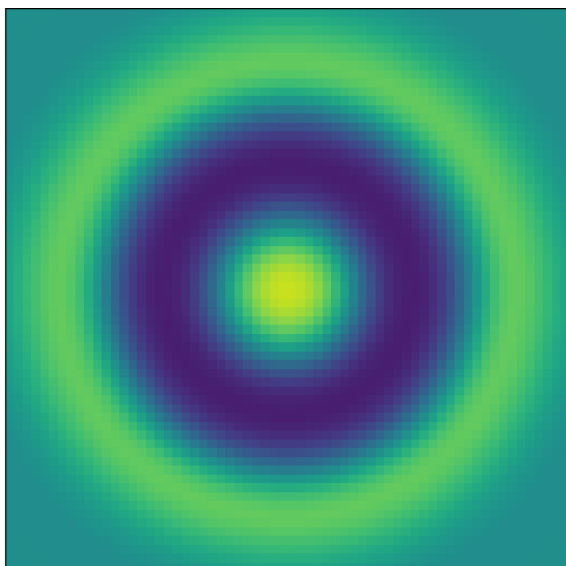
Non-defective system (spectral decomposition):

$$A = \underbrace{\begin{pmatrix} | & & | \\ p_1 & \cdots & p_n \\ | & & | \end{pmatrix}}_P \underbrace{\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}}_\Lambda \underbrace{\begin{pmatrix} - & p_1^{-1} & - \\ & \vdots & \\ - & p_n^{-1} & - \end{pmatrix}}_{P^{-1}}$$

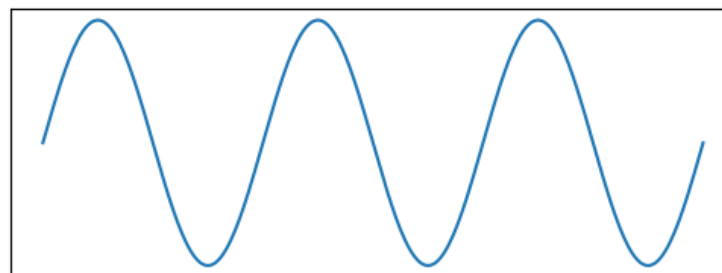
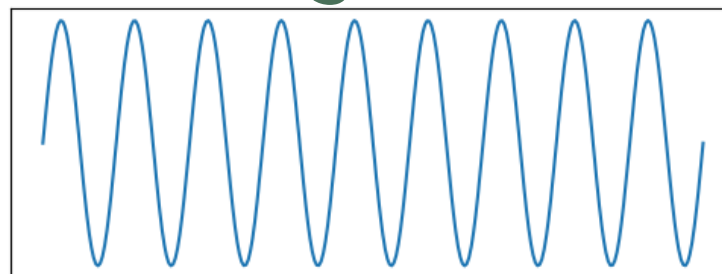
(Superposition principle)
$$x(t) = \sum_{k=1}^n \boxed{e^{t\lambda_k}} \boxed{p_k \langle p_k^{-1}, x_0 \rangle} = \sum_{k=1}^n \boxed{e^{t\tau_k + i2\pi\omega_k t}} \boxed{m_k(x_0)}$$

➔ Sum of elementary solutions that decouple **temporal** and **static** components.

Modes



Signals



- $\tau_k = \mathcal{R}(\lambda_k)$ timescale.
- $\omega_k = \mathcal{I}(\lambda_k)/2\pi$ frequency.
- $m_k(x_0) = \langle p_k^{-1}, x_0 \rangle p_k$ **mode**.

Non-linear autonomous dynamical systems

Modeling:

$$\dot{x}(t) = g(x(t)) \quad \rightarrow \quad x(t) = S_t(x_0) = x_0 + \int_0^t g(x(\tau)) d\tau$$

Koopman operator: Suppose a function space $\mathcal{F} \subset \mathbb{R}^{\mathcal{X}}$ (observables), for any time $t > 0$, the Koopman operator $T_t : \mathcal{F} \rightarrow \mathcal{F}$ associated to the dynamic verifies:

$$[T_t f](x_0) \triangleq f(S_t(x_0)) \quad \forall f \in \mathcal{F}, \forall x_0 \in \mathcal{X}$$

Remarks: Koopman operators are linear: $T_t(f + cg) = T_t f + cT_t g$

Infinitesimal generator: The semi-group of operators $\{T_t\}_{t>0}$ (i.e. $T_{s+t} = T_s T_t$) admits an infinitesimal generator such that:

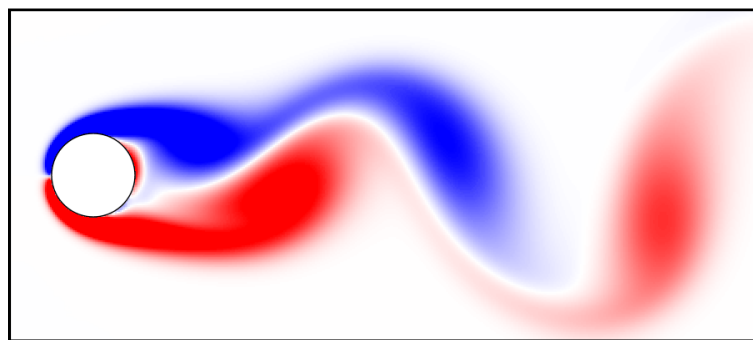
$$A = \lim_{t \rightarrow 0} (T_t - Id)/t \quad \text{and} \quad T_t = \exp(tA)$$

Operator spectral decomposition

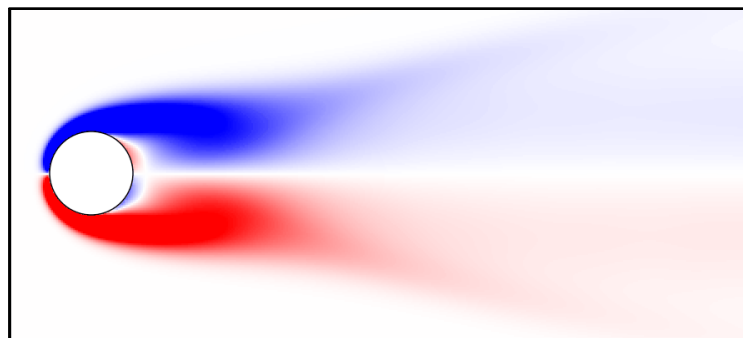
Spectral decomposition: Assuming that $\mathcal{F}$ is a separable Hilbert space and A is a non defective operator with purely discrete spectrum. Then A can be written as:

$$Af = \sum_{j \in \mathbb{N}} \lambda_j \langle g_j, f \rangle f_j \quad \text{s.t.} \quad Lf_j = \lambda_j f_j, \quad L^*g_j = \bar{\lambda}_j g_j, \quad \langle f_i, g_j \rangle = \delta_{i,j}$$

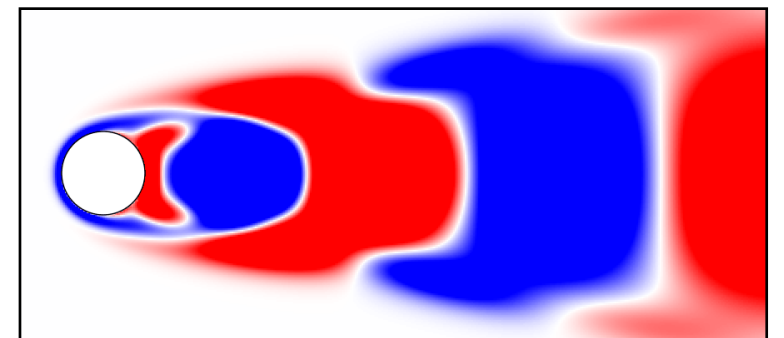
Raw signal



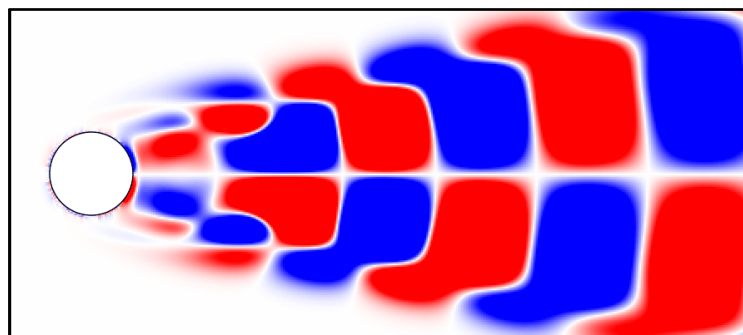
Mode 1 (0Hz)



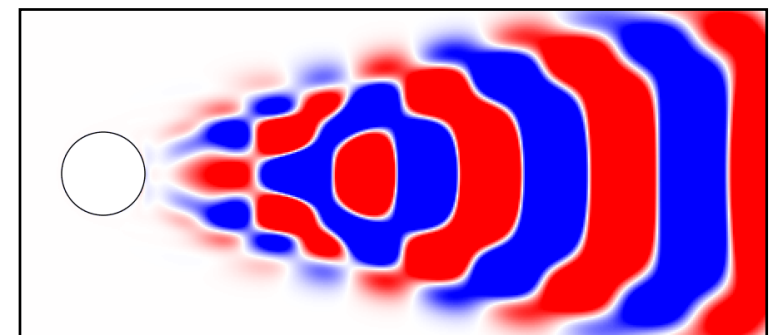
Mode 2 (3.3Hz)



Mode 3 (6.4Hz)



Mode 4 (9.3Hz)



Time Continuous Markov Process

Setting: Let assume a continuous-time Markov process $(X_t)_{t \geq 0} \subseteq \mathcal{X}$ described by an family of probability densities $\{p_t(x, y)\}_{t \geq 0}$ such that:

$$P_t(x, A) = P(X_t \in A \mid X_0 = x) = \int_A p_t(x, y) dy ,$$

with an invariant measure π , i.e. $\pi P_t = \pi \quad \forall t \geq 0$.

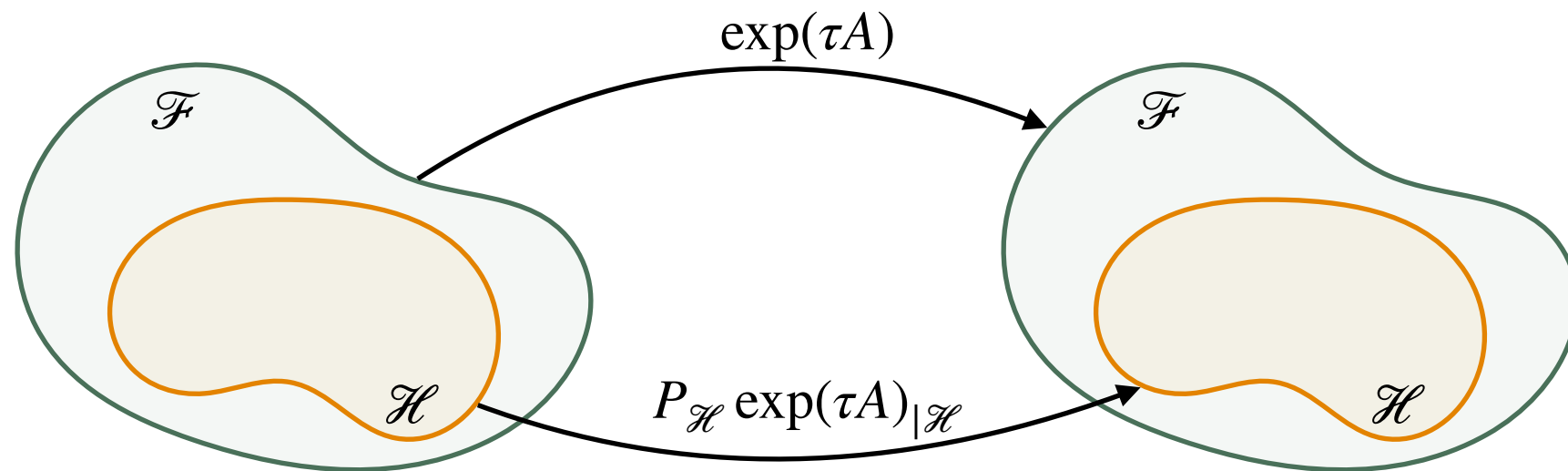
Transfert operators: Let $L_\pi^2(\mathcal{X})$ be the set of square integrable function with respect to π , we consider the transfert operators for any $t > 0$:

$$[A_t(f)](x) \triangleq \mathbb{E}[f(X_t) \mid X_0 = x], \quad x \in \mathcal{X} \text{ and } f \in L_\pi^2(\mathcal{X}),$$

and its infinitesimal generator $A = \lim_{t \rightarrow 0} (A_t - Id)/t$.

Learning transfer operators

Problem: Only trajectories of dynamical systems are observed. Neither the operator A nor its domain $\mathcal{F}$ are known.



Popular approach [Bru22, Kos22]: Learning the action of $A : \mathcal{F} \rightarrow \mathcal{F}$ on a predefined finite dimensional RKHS $\mathcal{H} \subset \mathcal{F}$ with basis $(f_i)_{i \in [d]}$ and features map $\phi(x) = (f_1(x), \dots, f_d(x))$ via empirical risk minimization over samples $\{x_t\}_{t \in [n]}$:

$$\operatorname{argmin}_{A \in \mathbb{R}^{d \times d}, \operatorname{rank}(A)=r} \frac{1}{n} \sum_{t \in [n]} \|z_{t+1} - A^\dagger z_t\|_{\mathcal{H}}^2 \quad \text{s.t.} \quad z_t = \phi(x_t)$$

Optimal Transport metric between operator representations

Spectral-Grassmann Optimal transport [Ger25]

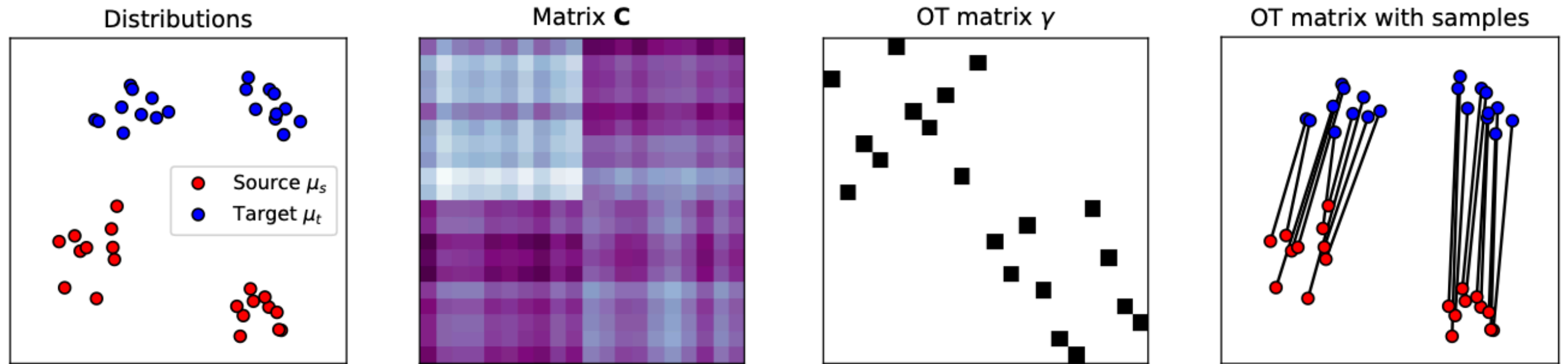
Considered space: Let $\mathcal{H}$ be a separable $\mathbb{C}$ -Hilbert space and $\mathcal{S}_r(\mathcal{H})$ the set of non-defective operator on $\mathcal{H}$ with rank at most $r \in \mathbb{N}$. For any $A \in \mathcal{S}_r(\mathcal{H})$:

$$Af = \sum_{i \in [\ell]} \sum_{j \in [m_i]} \lambda_i \langle g_{i,j}, f \rangle f_{i,j} = \sum_{i \in [\ell]} \lambda_i P_i f, \quad \text{s.t.} \quad \langle f_{i,j}, g_{u,v} \rangle = \delta_{(i,j),(u,v)}$$

Distributional embedding: The operator A can be represented as a probability distribution over the product space of eigenvalues and projectors:

$$\mu(A) \triangleq \sum_{i \in [\ell]} \frac{m_i}{m} \delta_{(\lambda_i, P_i)} \quad \text{s.t.} \quad m = \sum_{i \in [\ell]} m_i < r .$$

Discrete Optimal Transport



Kantorovich formulation:

Given two probability distributions $\mu_s = \sum_{i \in [n_s]} a_i \delta_{x_i^s}$ and $\mu_t = \sum_{i \in [n_t]} b_i \delta_{x_i^t}$, the optimal transport cost of order $p \in \mathbb{N}^*$ verifies:

$$W_{d,p}^p(\mu_s, \mu_t) = \min_{P \in \Pi(\mu_s, \mu_t)} \langle C, P \rangle_F,$$

where C is the cost matrix $c_{ij} = d(x_i^s, x_j^t)^p$, and the transport matrix P is from the set:

$$\Pi(\mu_s, \mu_t) = \{P \in \mathbb{R}_+^{n_s \times n_t} \mid P1_{n_t} = a, P^\top 1_{n_s} = b\}$$

Remarks:

- Computable with network simplex in $n^3 \log(n)$ where $n = n_s = n_t$.
- $W_d^p(\mu_s, \mu_t)$ is called a Wasserstein metric.

Metric space definition

Non-defective Optimal Transport metric: Let $\mathcal{H}$ be a separable $\mathbb{C}$ -Hilbert space and $\mathcal{S}_r(\mathcal{H})$ the set of non-defective operator on $\mathcal{H}$ with rank at most $r \in \mathbb{N}$. For $p \geq 1$ and $\eta \in (0,1)$, the Spectral-Grassmann Wasserstein metric distance between two operators $A, A' \in \mathcal{S}_r(\mathcal{H})$:

$$d_{\mathcal{S}}(A, A') \triangleq W_{d_{\eta}, p}(\mu(A), \mu(A')),$$

where the cost function is defined by the metric:

$$d_{\eta}((\lambda, P), (\lambda', P')) \triangleq \eta |\lambda - \lambda'| + (1 - \eta) d_{\mathcal{G}}(P, P').$$

Then $(\mathcal{S}_r(\mathcal{H}), d_{\eta})$ is a metric space.

Example of $d_{\mathcal{G}}$: $\|P - P'\|_{HS}$ or $-\log(|\langle P_i, P'_j \rangle| / (\|P_i\| \|P'_j\|))$

Computation:

- Pre-compute the cost matrix C with complexity $\mathcal{O}(r^2 n^2)$.
- Solve the OT problem with complexity $\mathcal{O}(r^3 \log(r))$ with network simplex.
- Overall complexity $\mathcal{O}(r^2 n^2 + r^3 \log(r))$.

Metric estimator's converge bound

Recall: Under kernel universality assumption such methods come with spectral estimation guarantees [Kos23], specifically for the Reduced Rank Regression (RRR) method [Kos22].

Theorem (simplified): Suppose two dynamical systems with low rank operator projection $A, A' \in \mathcal{S}_r(\mathcal{H})$ and $\widehat{A}, \widehat{A}'$ their estimation from n samples with the RRR method [Kos22]. Suppose $\alpha \in (1,2)$, and $\beta \in [0,1]$ bounding the empirical covariance. In the i.i.d setting, for any $\delta \in (0,1)$, with probability $1 - \delta$ it holds:

$$|d_{\mathcal{S}}(\widehat{A}, \widehat{A}') - d_{\mathcal{S}}(A, A')| \leq Cn^{-\frac{\alpha-1}{2(\alpha+\beta)}} \ln(2\delta^{-1})$$

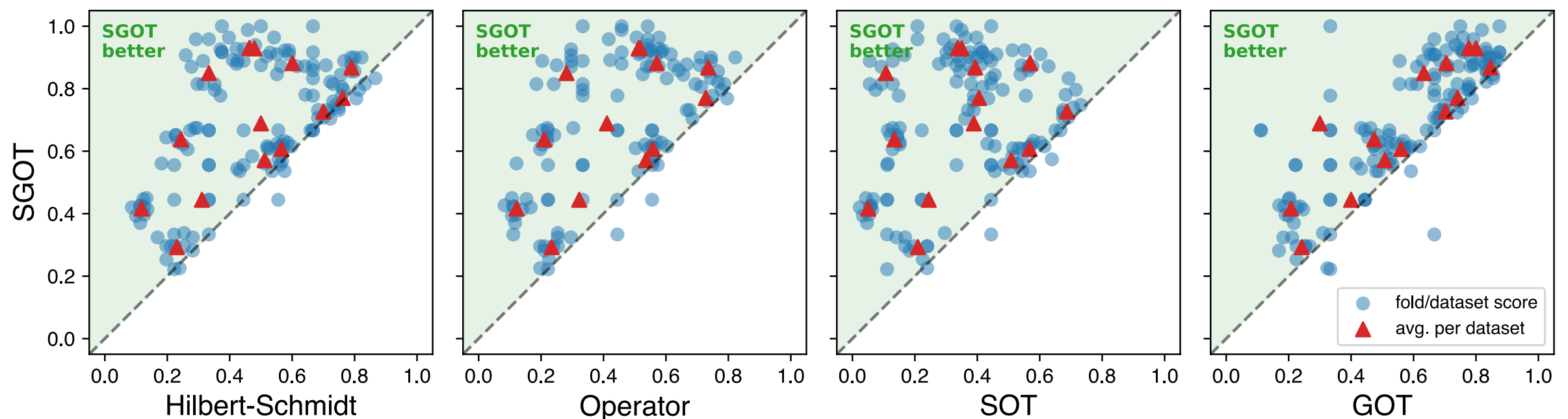
Classification

Dataset: 14 datasets from the UEA time series database.

Classifier: K-nearest-neighbors with 6 different metrics.

Score function: Accuracy.

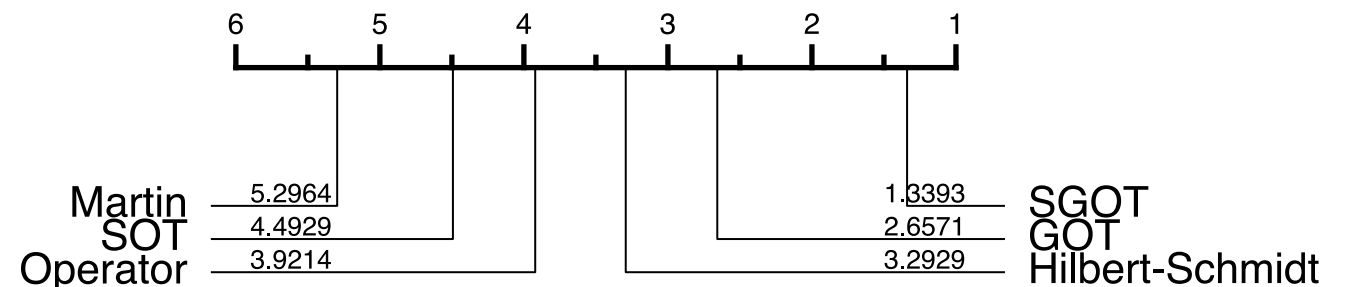
Protocol: Nested cross-validation + grid search for neighbors K and SGOT ratio η .



Avg. computation time

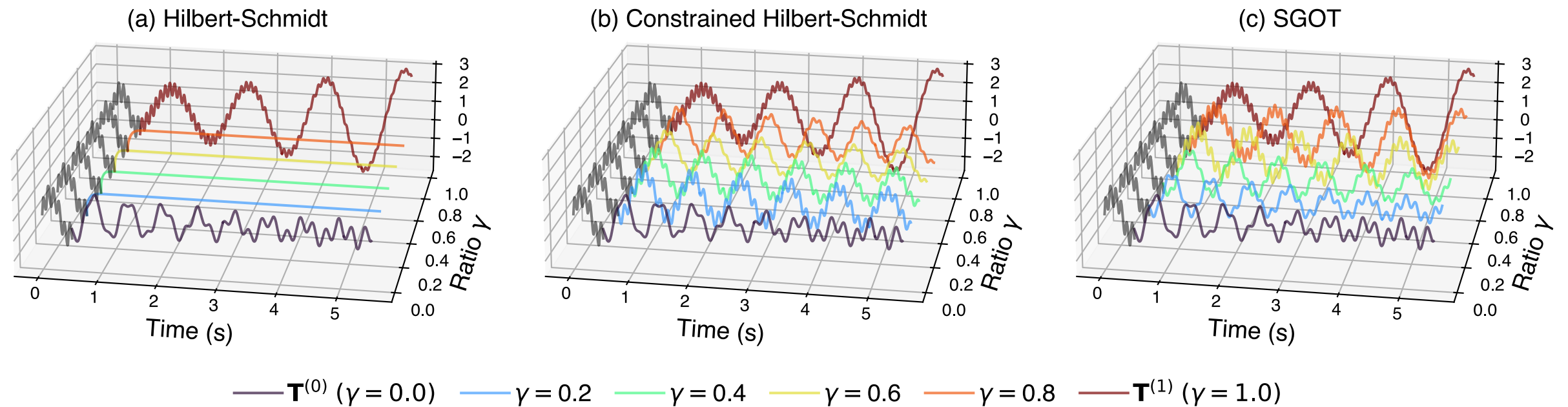
Hilbert-Schmidt	Operator	Martin	SOT	GOT	SGOT
4.96ms	13.04ms	0.02ms	0.03ms	0.14ms	0.12ms

Critical Diagram difference

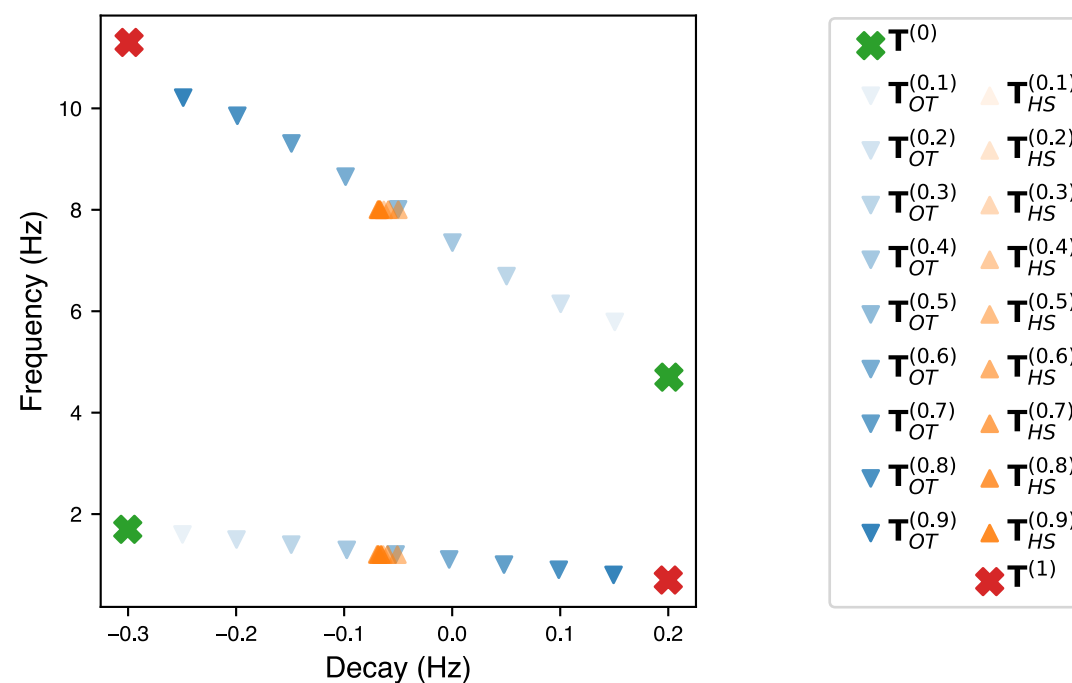


Barycenter & Interpolation

Barycentric interpolation between two linear dynamical systems.

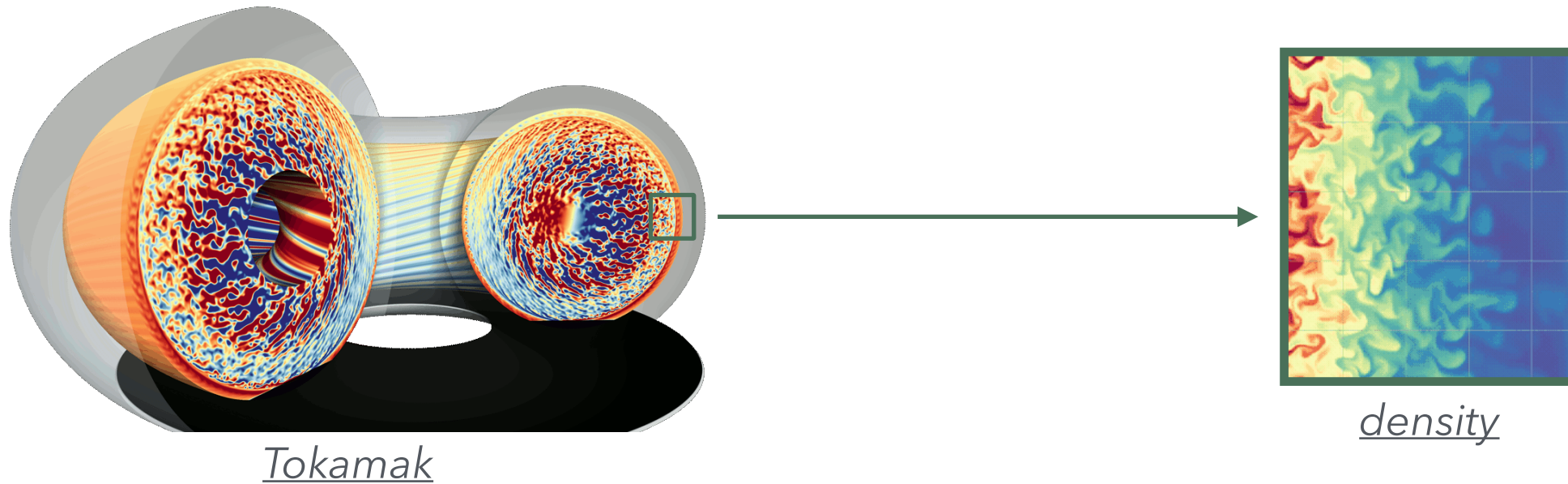


Barycenters' eigenvalues evolution:



Dictionary Learning for Dynamical Systems

Application: Plasma dynamic identification



Challenge: Heat and confine plasma long enough for fusion reactions.

Plasma PDE model:

$$\begin{cases} \partial_t n + \kappa \partial_y \phi + [\phi, n] - D_n \Delta_{\perp} n = -\sigma_n n + \sigma_{n\phi} \phi, \\ \partial_t W + g \partial_y n + [\phi, W] - D_{\phi} \Delta_{\perp} W = -\sigma_{\phi n} n + \sigma_{\phi\phi} \phi \\ W = \Delta_{\perp} \phi \end{cases}$$

with n the particle density, ϕ the electric potential and W the vorticity.

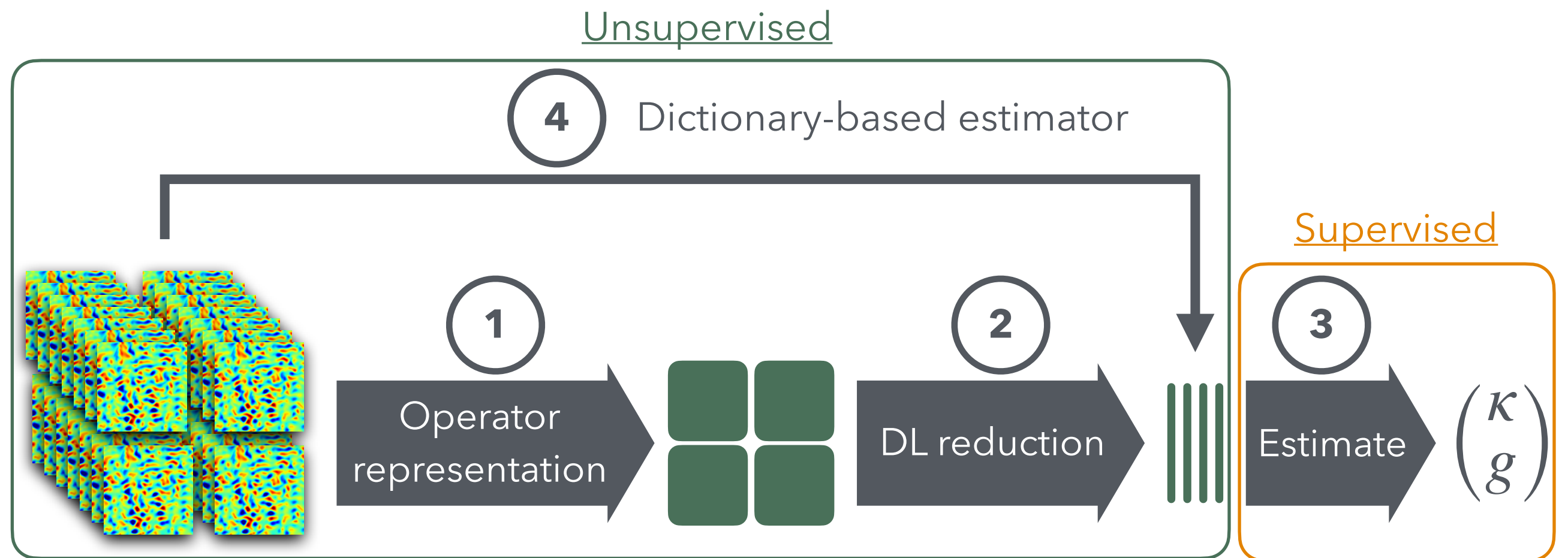
Intermediate challenge:

Can we estimate parameters κ and g ? Even from short trajectories ?

Overview

Developed methodology:

1. A pipeline to transform raw data into PDE's operator representation (matrix)
2. Compress representation with dictionary learning (DL) (matrix to small vector)
3. Estimation of PDE's parameters from compressed representations.
4. Dictionary-based estimation of operator representations.



Dictionary learning embedding reduction



Main question:

How to compress operator representation ?

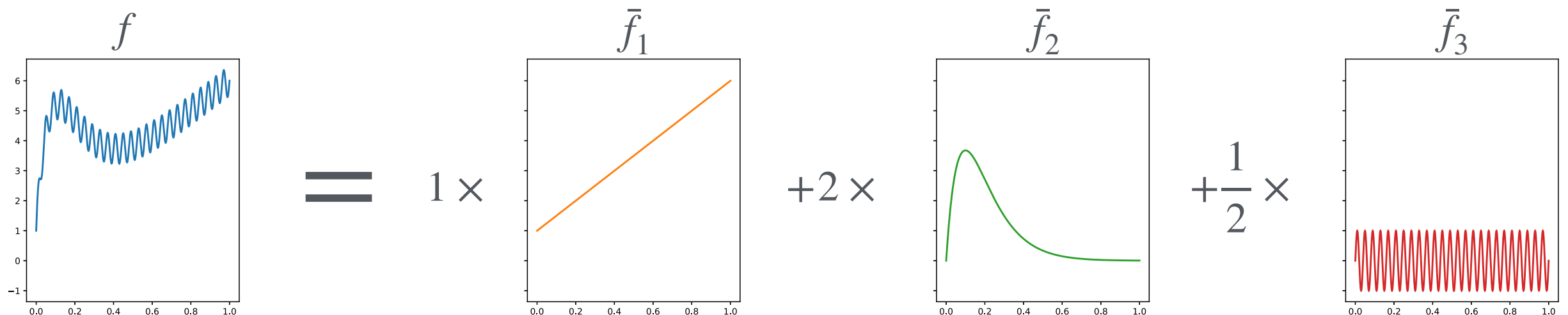
Dictionary learning

Core idea: Representing any object $X \in \mathcal{X}$ by its decomposition $\alpha \in \mathbb{R}^D$ in a dictionary of D objects $\{\bar{X}_i\} \in \mathcal{X}^D$ with a composition function $\mathbf{B}$, i.e. $X \approx \mathbf{B}(\alpha, \{\bar{X}_i\})$.

Training: Given a set of N objects $\{X_j\}$, the dictionary and the encoding vectors are typically learned by minimizing a reconstruction error :

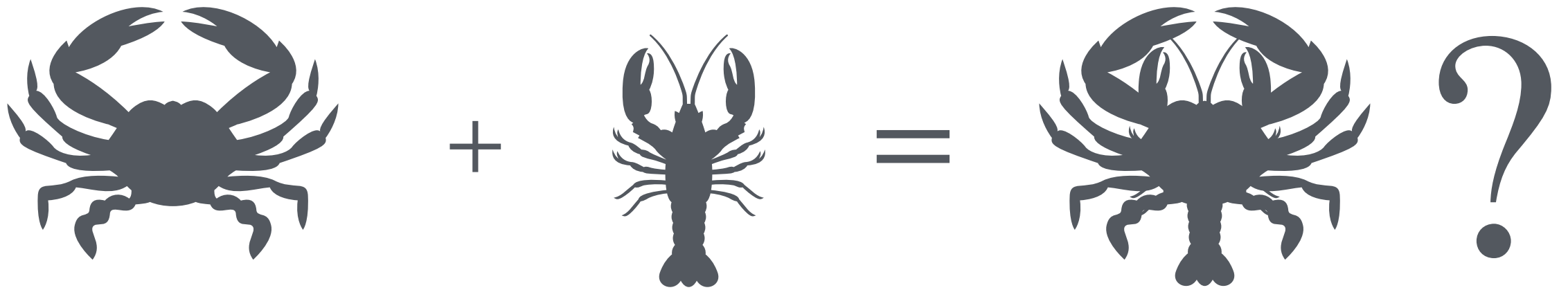
$$\min_{\{\alpha_j\}_{j \in [N]}, \{\bar{X}_i\}_{i \in [D]}} \sum_{j \in [N]} \mathcal{L}(X_j, \mathbf{B}(\alpha_j, \{\bar{X}_i\}))$$

Example on functions:

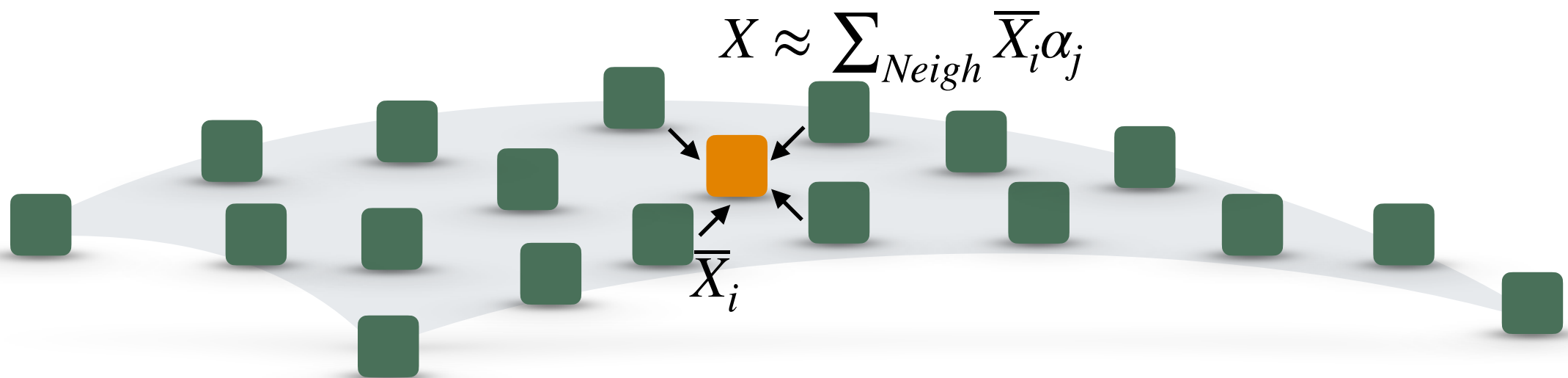


$$\min_{\Lambda, \mathbf{D}} \|\mathbf{X} - \mathbf{D}\Lambda\|_2^2 \quad \text{with} \quad \mathbf{X}_{ij} = f_j(t_i), \mathbf{D}_{ik} = \bar{f}_k(t_i) \text{ and } \Lambda_{k,j} = \alpha_{kj}$$

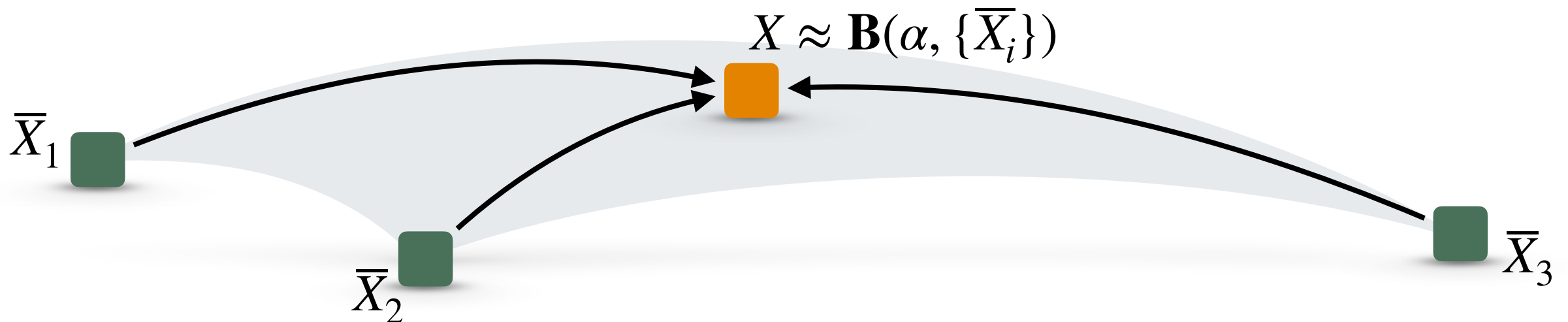
Dictionary learning in non-Euclidean space



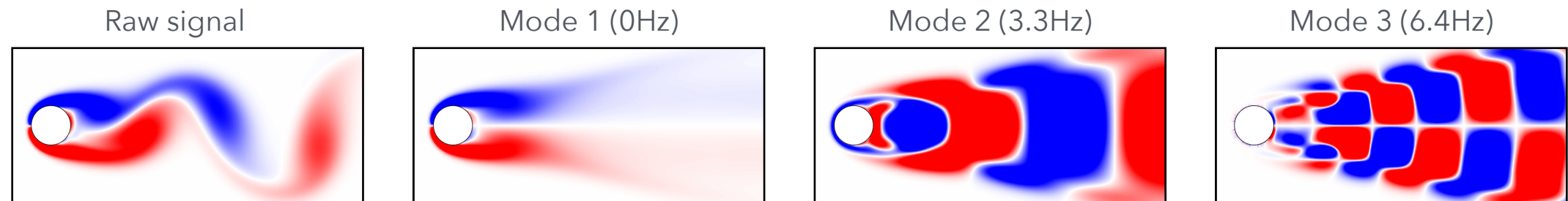
1st strategy: Overcomplete (Large dictionary with sparse coding).



2nd strategy: Incorporating an apriori of the dataset manifold in modeling.



Dictionary learning on operator of DS



Geometry of spectral decomposition: Let $\mathcal{H}$ be a p -dimensional Hilbert space and $\mathcal{S}_r(\mathcal{H})$ be the set of non-defective operators with rank at most r and simple spectrum. For any $\mathbf{A} \in \mathcal{S}_r(\mathcal{H})$:

$$\mathbf{A} = \mathbf{R} \text{Diag}(\Lambda) \mathbf{L}^\dagger \quad \text{s.t.} \quad \mathbf{L}^\dagger \mathbf{R} = \mathbf{I}_r,$$

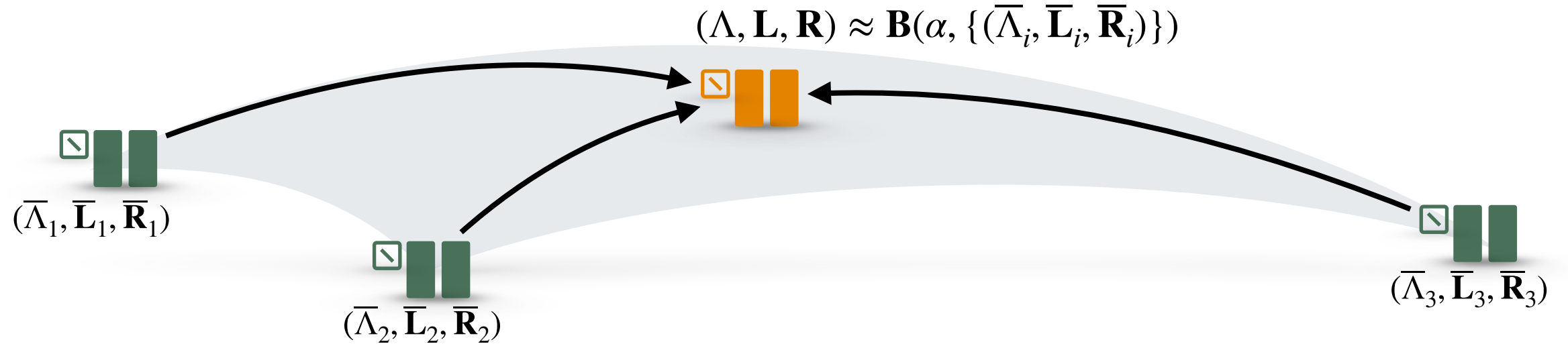
with eigenvalues $\Lambda \in \mathbb{C}^r$ and left/right eigenvectors $(\mathbf{L}, \mathbf{R}) \in (\mathbb{C}^{p \times r})^2$.

⚠ A spectral decomposition uniquely defines an operator up to change of inner basis. Hence, $\mathcal{S}_r(\mathcal{H})$ can be identified with:

$$\mathcal{M} = \mathcal{N} / (GL_r(\mathbb{C})) \quad \text{s.t.} \quad \mathcal{N} = \{(\Lambda, \mathbf{L}, \mathbf{R}) \mid \mathbf{L}^\dagger \mathbf{R} = \mathbf{I}_r\}$$

Proposition: $(\mathcal{M}, g)$ is a Riemannian quotient manifold of $(\mathcal{N}, g)$ where g is the stable Riemannian metric.

Dictionary learning on operator of DS



Objective: Given a set of N operators with decomposition $\{(\Lambda_j, \mathbf{L}_j, \mathbf{R}_j)\}$, the dictionary of D atoms and the encoding vectors are learned with:

$$\min_{\{\alpha_j\}, \{(\bar{\Lambda}_i, \bar{\mathbf{L}}_i, \bar{\mathbf{R}}_i)\}} \sum_{j \in [N]} d_{\mathcal{S}}^2 \left((\Lambda_j, \mathbf{L}_j, \mathbf{R}_j), \mathbf{B}(\alpha_j, \{(\bar{\Lambda}_i, \bar{\mathbf{L}}_i, \bar{\mathbf{R}}_i)\}) \right)$$

with the composition function:

$$\begin{aligned} \mathbf{B}(\alpha_j, \{(\bar{\Lambda}_i, \bar{\mathbf{L}}_i, \bar{\mathbf{R}}_i)\}) &= \mathbf{P}_{\mathcal{N}} \left(\sum_i \alpha_{ji} \bar{\Lambda}_i, \sum_i \alpha_{ji} \bar{\mathbf{L}}_i, \sum_i \alpha_{ji} \bar{\mathbf{R}}_i \right) \\ \text{s.t.} \quad \mathbf{P}_{\mathcal{N}}(\Lambda, \mathbf{L}, \mathbf{R}) &= (\Lambda, \arg \min_{\tilde{\mathbf{L}}} \|\tilde{\mathbf{L}} - \mathbf{L}\|_F, \mathbf{R}) \end{aligned}$$

Optimization: Stochastic inexact block-coordinate strategy

1. Coefficient estimation: complete gradient descent
2. Dictionary update: One Riemannian gradient descent iteration

Spectral-Grassmann Optimal transport [Ger25]

Spectral decomposition: For any $A \in \mathcal{S}_r(\mathcal{H})$,

$$Af = \sum_{j \in [r]} \lambda_j \langle g_j, f \rangle f_j = \sum_{j \in [r]} \lambda_j P_j f \quad \text{s.t.} \quad P_j f = \langle g_j, f \rangle f_j$$

with $A := (\Lambda, \mathbf{L}, \mathbf{R})$ then $\lambda_i := \Lambda_i$ and $P_i = \mathbf{R}_i \mathbf{L}_i^\dagger$

Distributional embedding: The operator A can be represented as a probability distribution over the product space of eigenvalues and projectors:

$$\mu(A) \triangleq \sum_{i \in [r]} \frac{1}{r} \delta_{(\lambda_i, P_i)} .$$

Metric between operator: Given two operator A and A' of rank r and r' , $d_{\mathcal{S}}$ verifies:

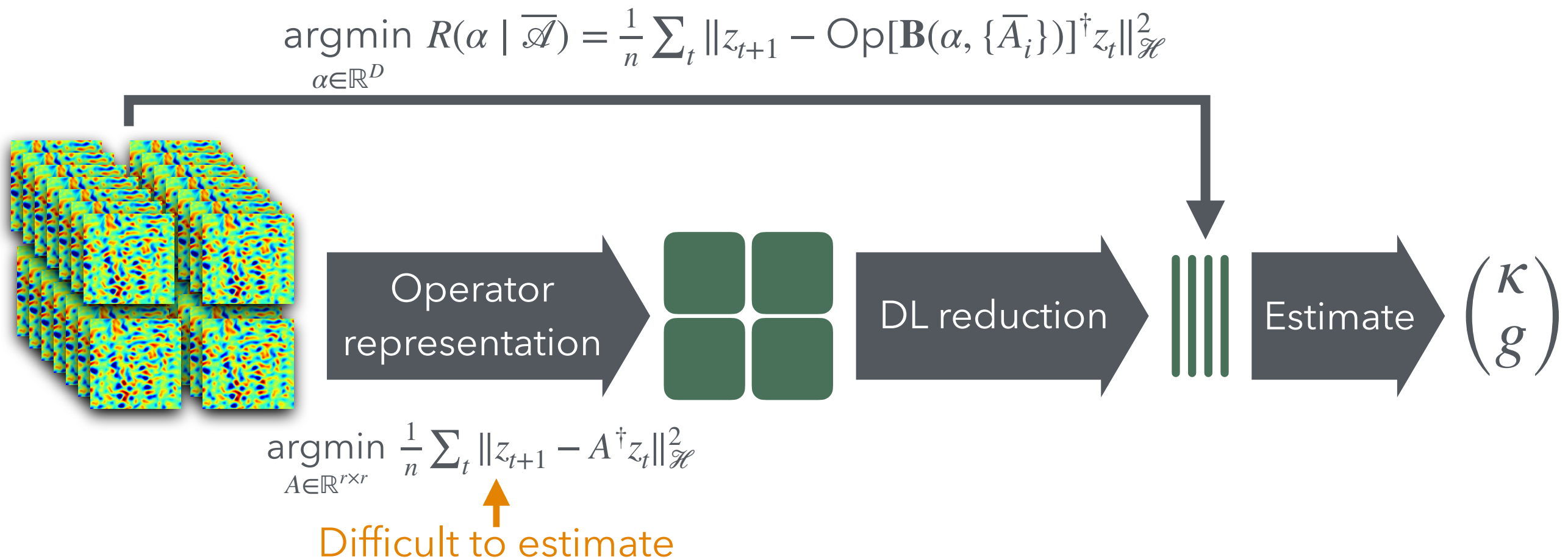
$$d_{\mathcal{S}}(A, A')^2 = \min_{\mathbf{P} \in \Pi} \langle \mathbf{C}, \mathbf{P} \rangle_F ,$$

where $\Pi = \{ \mathbf{P} \in \mathbb{R}_+^{r \times r'} \mid \mathbf{P} \mathbf{1}_{r'} = \mathbf{1}_r, \mathbf{P}^\top \mathbf{1}_r = \mathbf{1}_{r'} \}$, and

$$\mathbf{C}_{ij} = \eta |\lambda_i - \lambda'_j|^2 - (1 - \eta) \log(|\langle P_i, P'_j \rangle| / (\|P_i\| \|P'_j\|) + \epsilon)$$

Dictionary-based operator estimates

Problematic: How to estimate an operator from a short trajectory ?

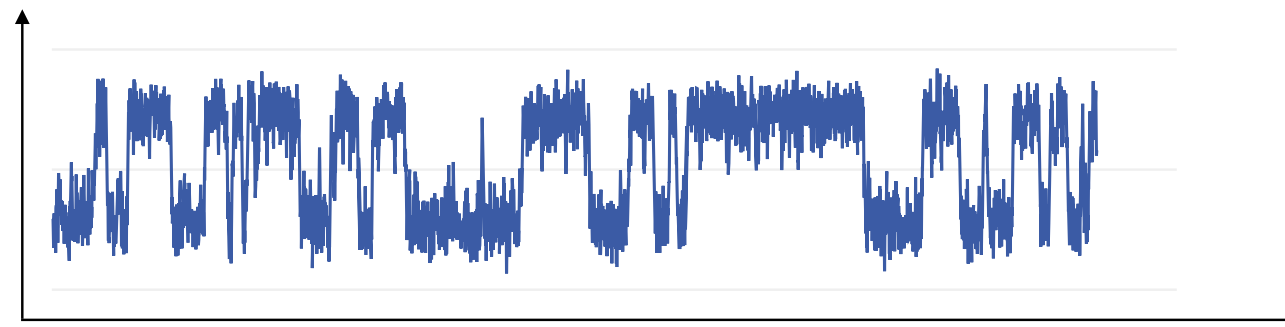
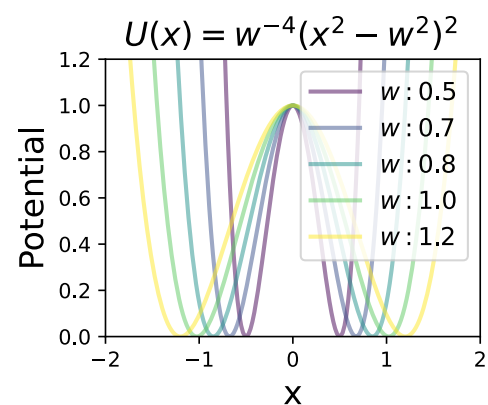


Theorem (Oracle inequality, i.i.d. case): Let $\overline{\mathcal{A}}$ be a dictionary with d atoms and $\mathcal{A}^*$ be its optimal counterpart. Suppose $\{X_t\}_{t \in [n]}$ are i.i.d. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$:

$$R(\hat{\alpha} \mid \overline{\mathcal{A}}) - \inf_{\alpha} R(\alpha \mid \mathcal{A}^*) \leq \text{bias}(\overline{\mathcal{A}}) + \sqrt{\text{Const}(\overline{\mathcal{A}}, \delta, \log(n))/n}$$

Experimental results

Langevin dynamic

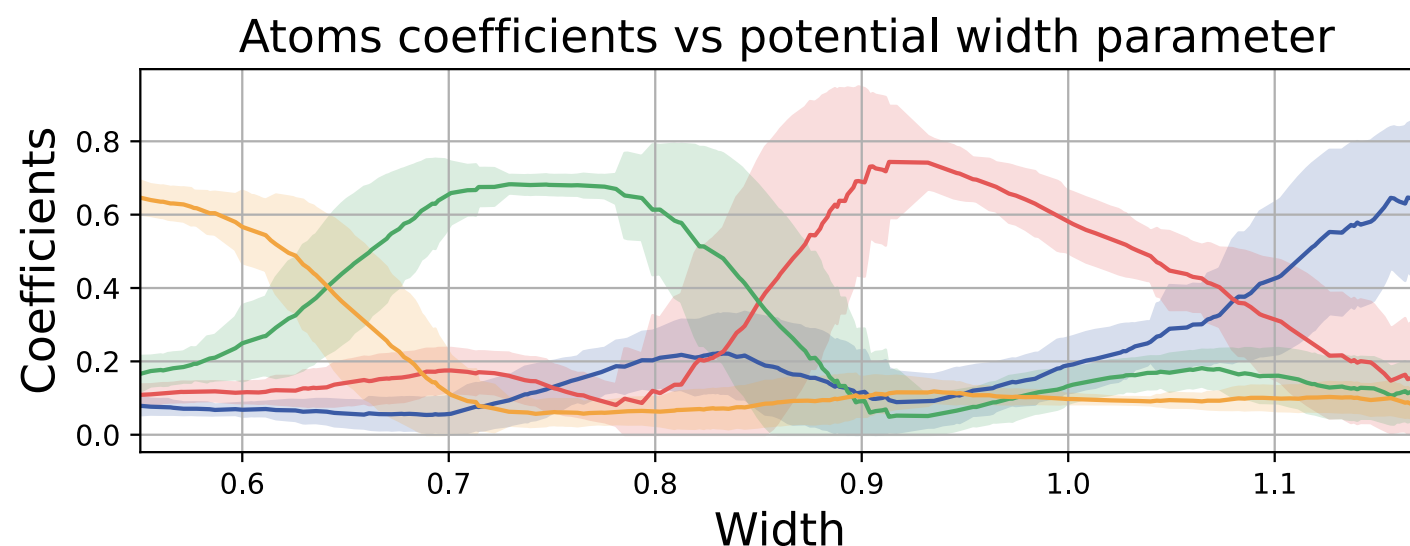
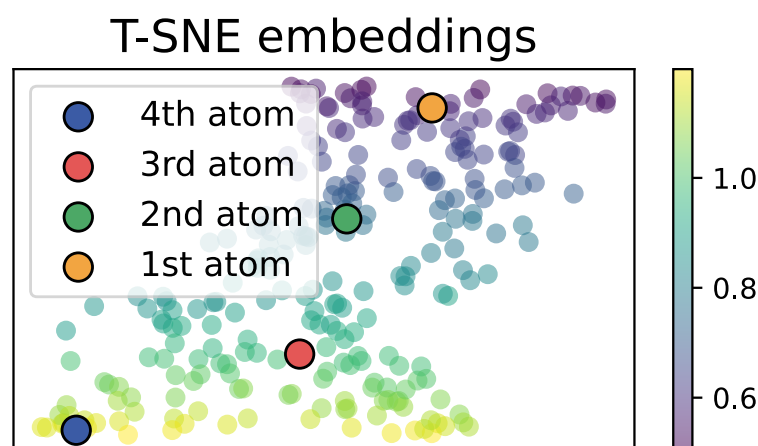


Modeling: (Damped Langevin)

$$dX_t = -\nabla U_w(X_t) + \sqrt{2\sigma}dB_t \text{ with } U_w(x) = w^{-4}(x^2 - w^2)^2.$$

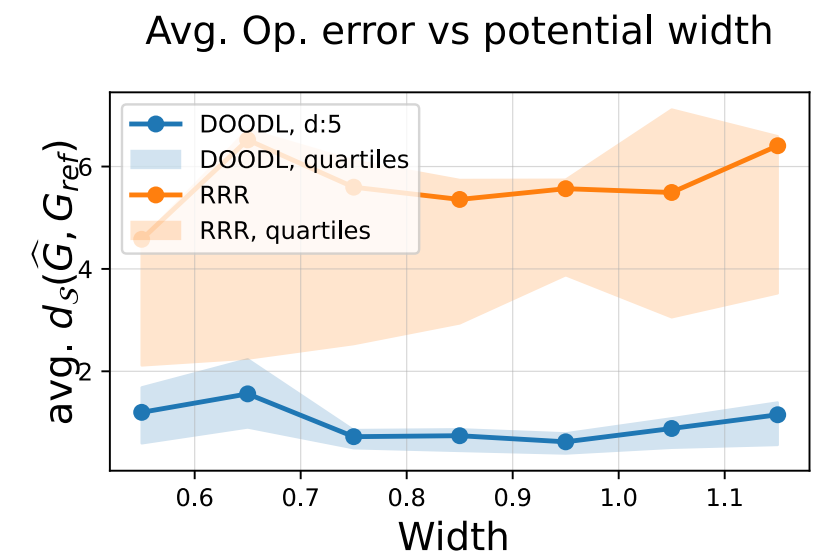
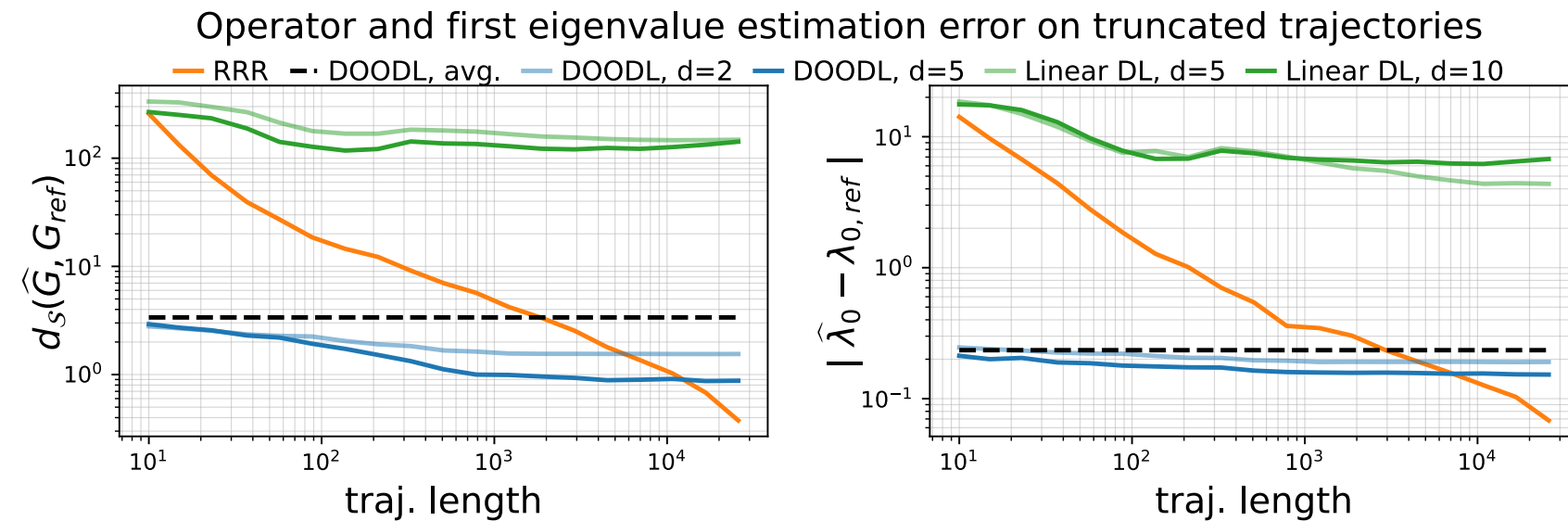
- 256/256 train/test trajectories with 40k samples each and $w \sim \text{Uniform}([0.5, 1.2])$
- Operator estimation: Reduced Rank Regression (RRR) [Kos22]
- Feature map: 400 Random Fourier Features (RFF) on 50 samples sliding windows.

Dictionary learning on train set:

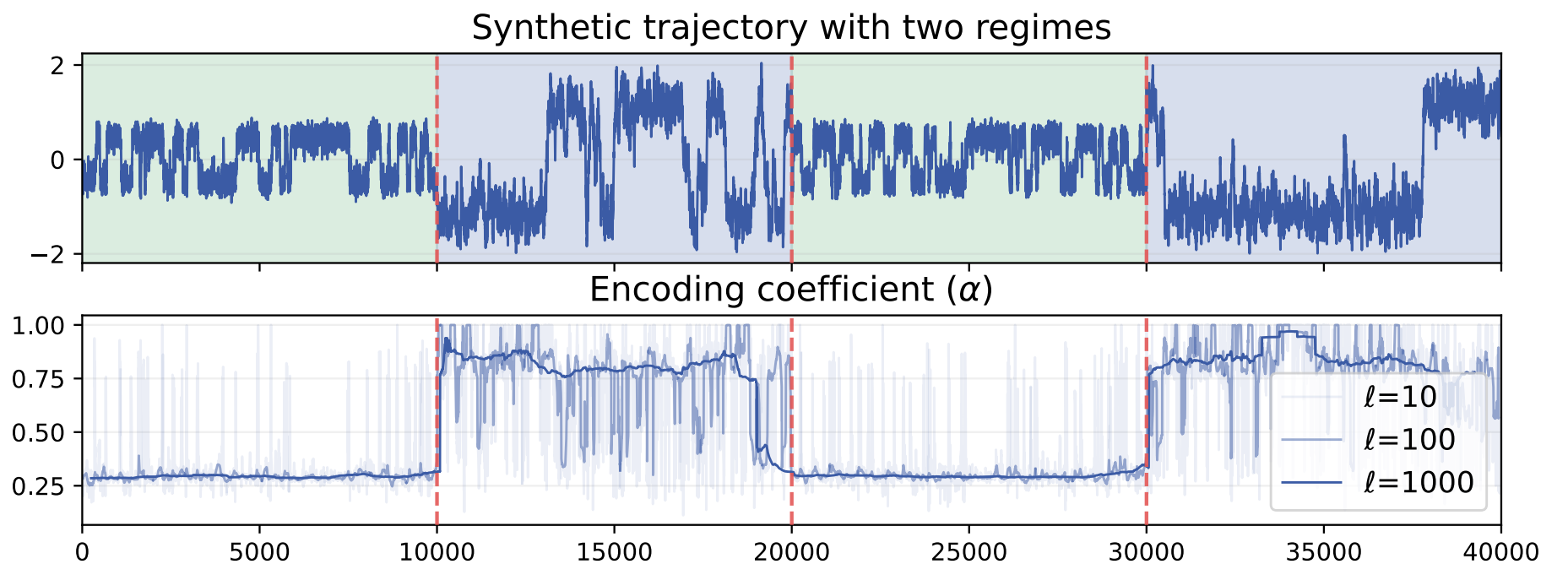
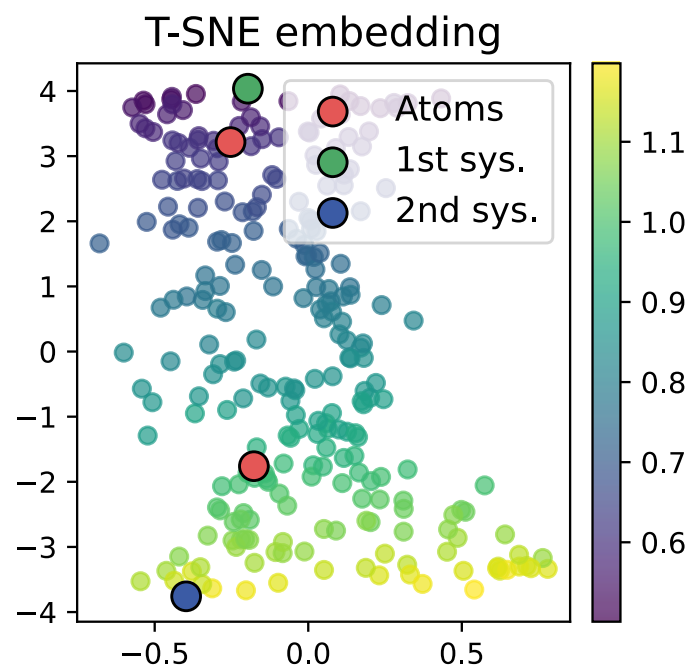


Langevin dynamic

Short trajectory operator estimation:



Switching dynamics estimation:



Plasma dataset generation

Plasma PDE model:

$$\begin{cases} \partial_t n + \kappa \partial_y \phi + [\phi, n] - D_n \Delta_{\perp} n = -\sigma_n n + \sigma_{n\phi} \phi, \\ \partial_t W + g \partial_y n + [\phi, W] - D_{\phi} \Delta_{\perp} W = -\sigma_{\phi n} n + \sigma_{\phi\phi} \phi \\ W = \Delta_{\perp} \phi \end{cases}$$

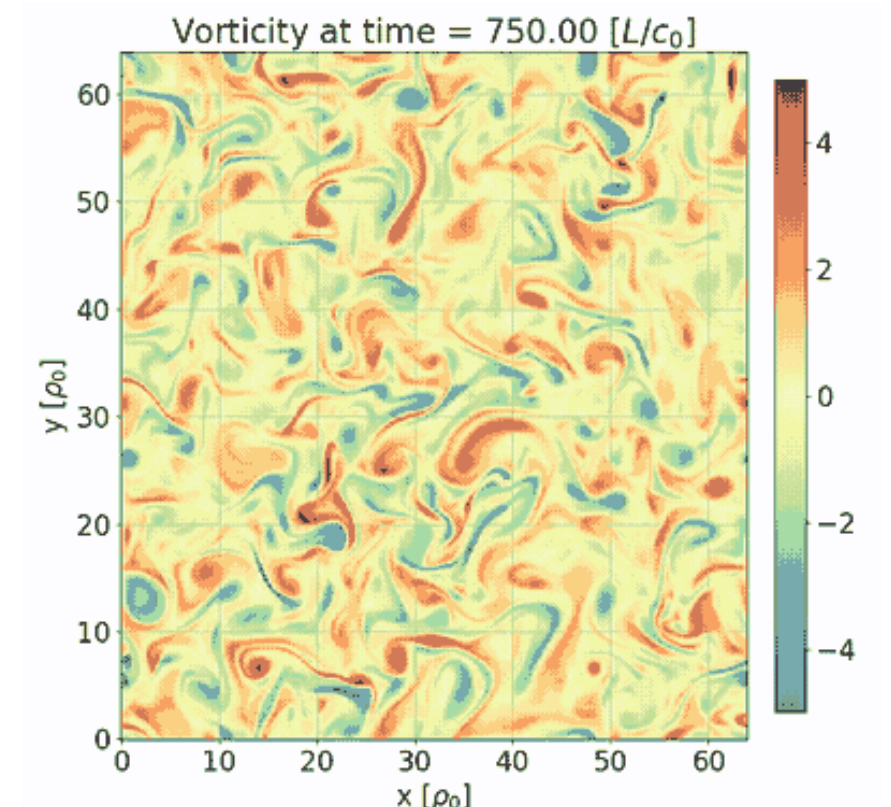
with n the particle density, ϕ the electric potential and W the vorticity.

Goal: Estimation of parameters κ and g , even from short trajectories.

Simulator: Tokam2D [Var23]

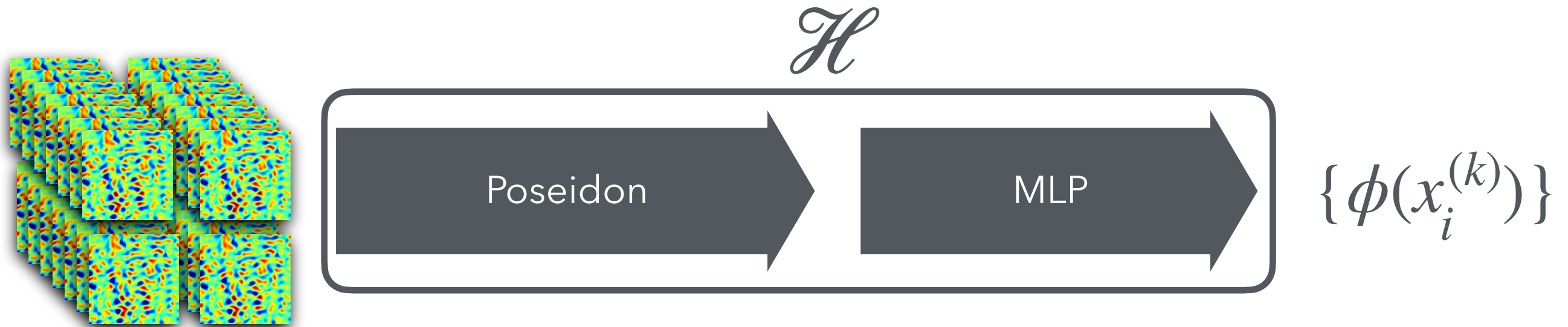
Data generation:

- 320/80 train/test trajectories.
- 128x128 grid, $\Delta t = 0.05s$, $n_{\text{samples}} = 4093$
- $(\kappa, g) \sim \text{Uniform}([1, 3.5] \times [0, 0.5])$
- Laplace transform operator estimator [Kos24]



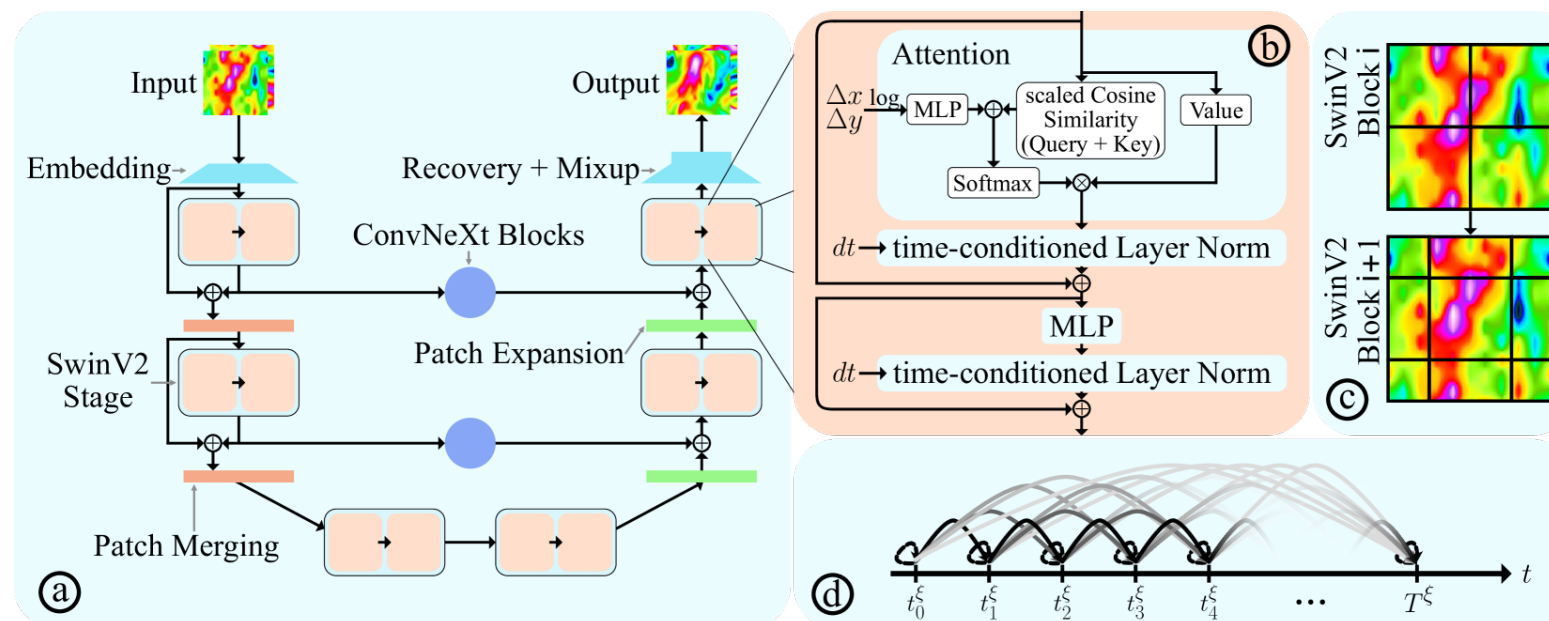
Observable space

Context: Learning a function space $\mathcal{H}$ common for the proper estimation of the whole family of systems $\{A^{(k)}\}_{k \in [K]}$.



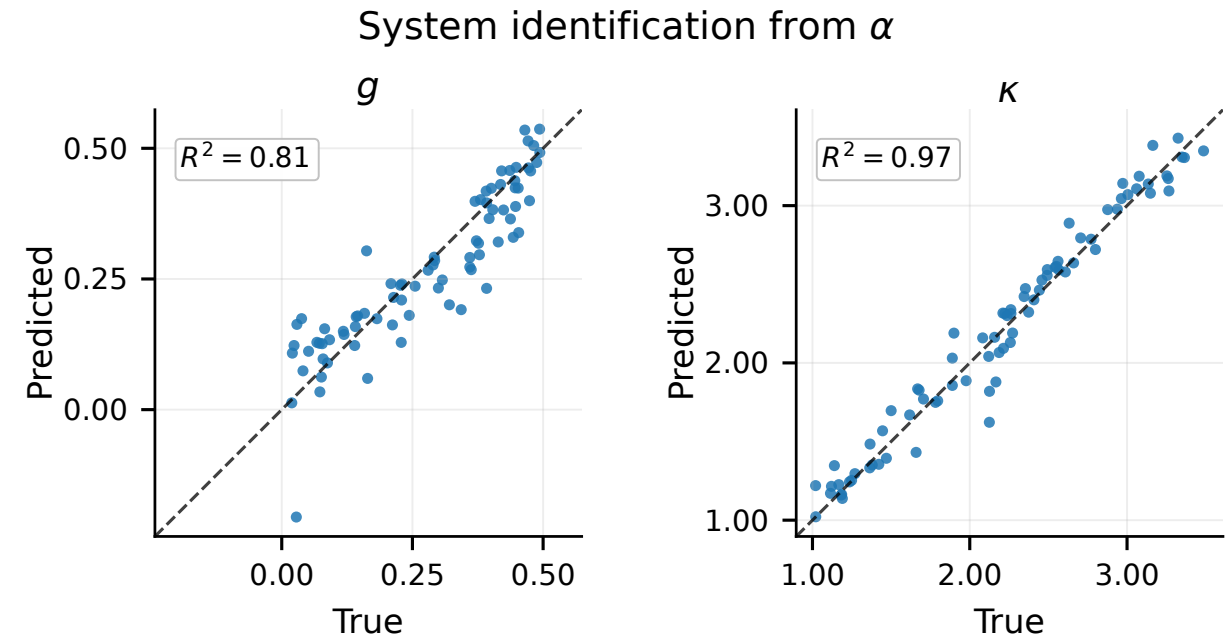
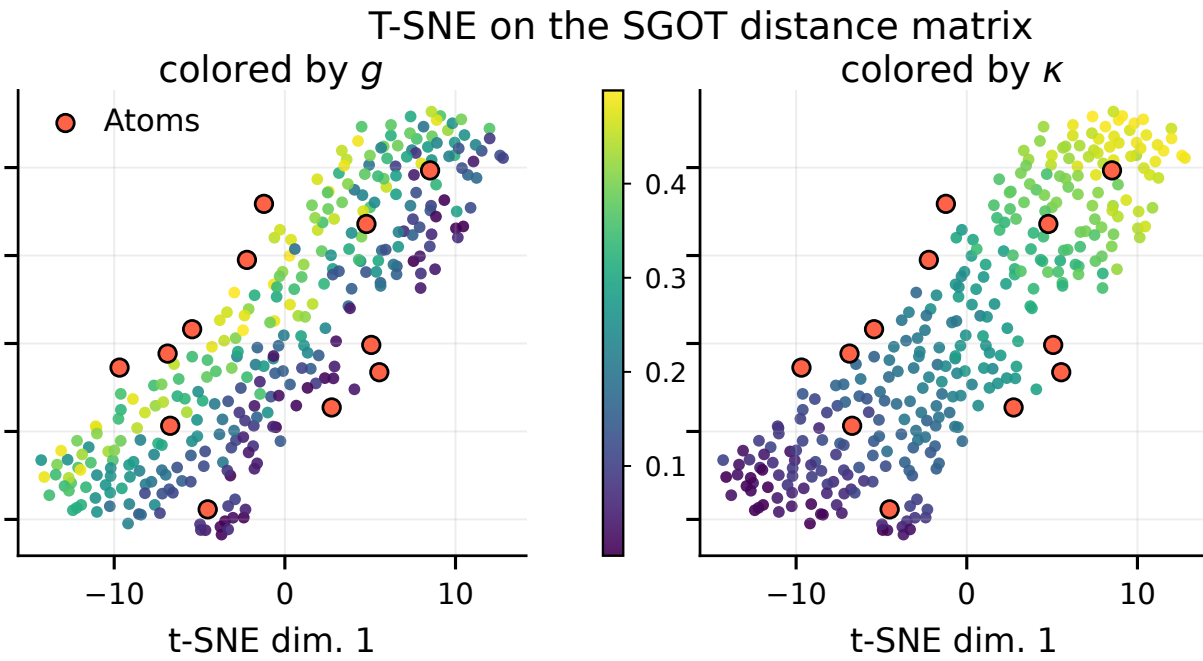
Training: Poseidon frozen, Fine tuning the MLP following [Kos24].

Poseidon: Foundation model for fluid dynamics (transformer based).

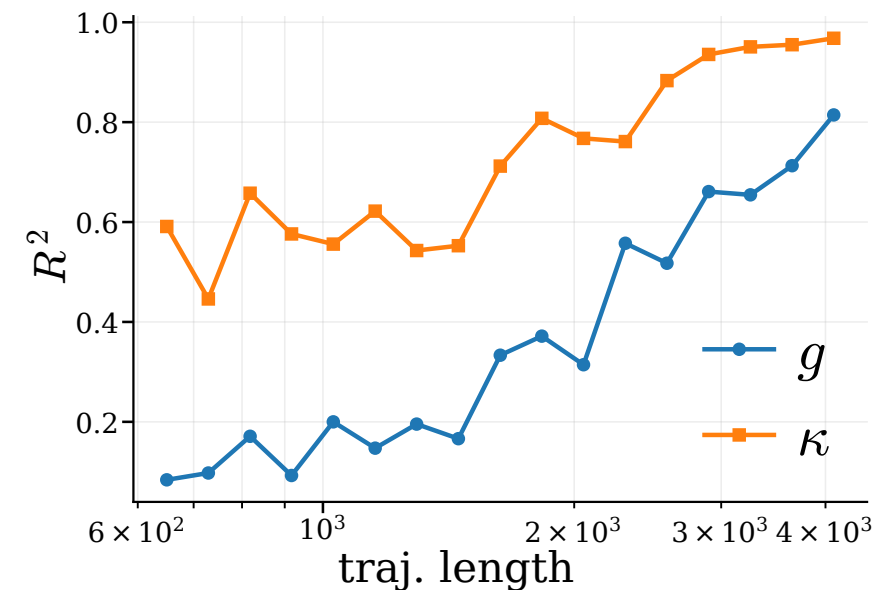
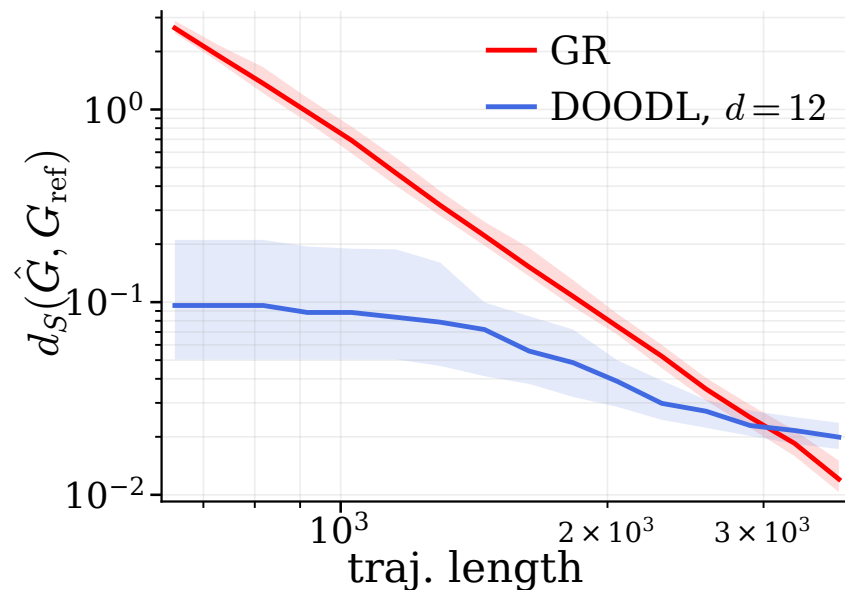


Dictionary learning & Parameter estimation

Full trajectory:



Small to full trajectory:



Thank you !
Any questions ?